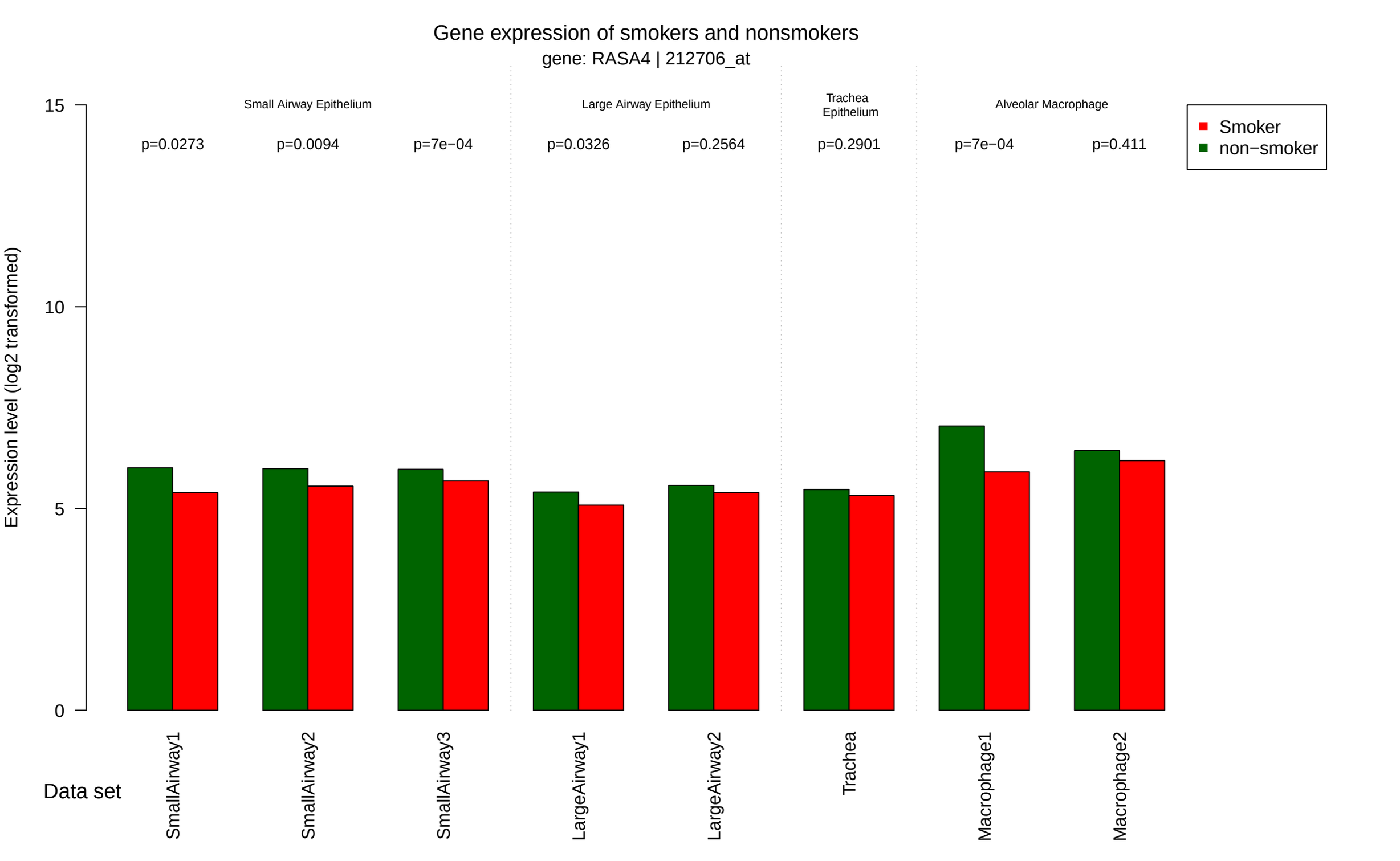
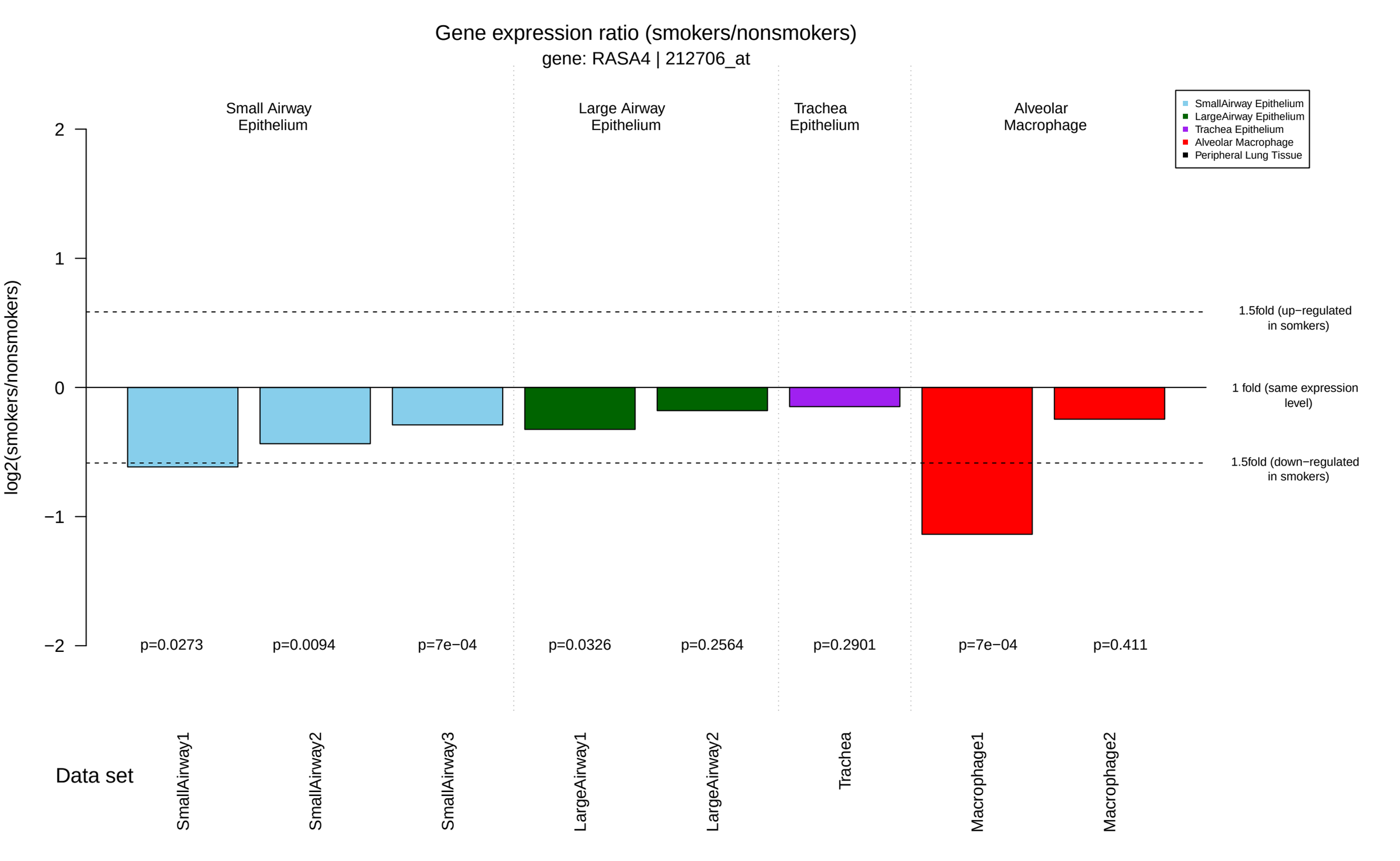
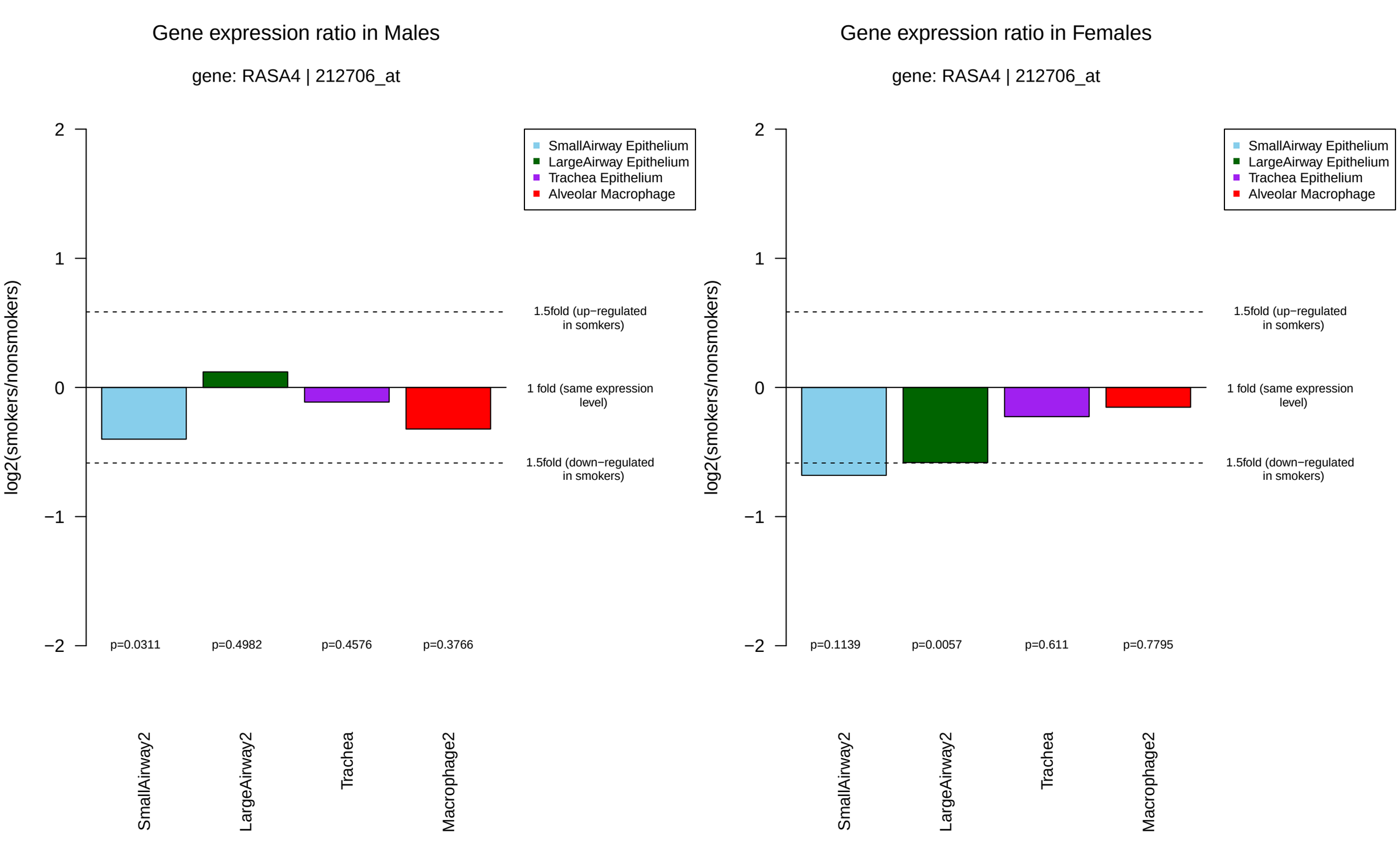
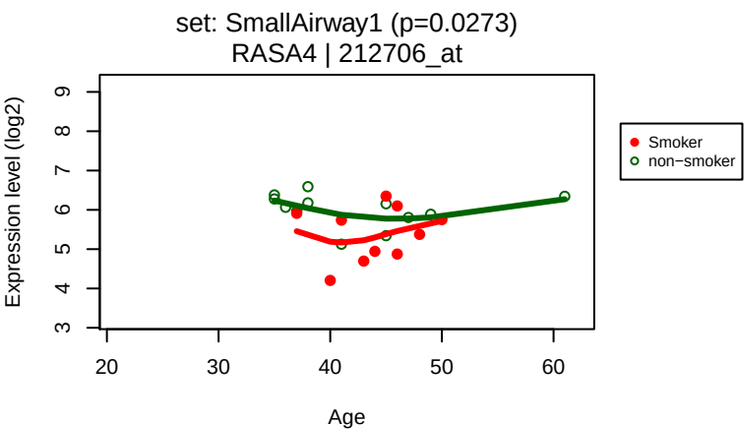
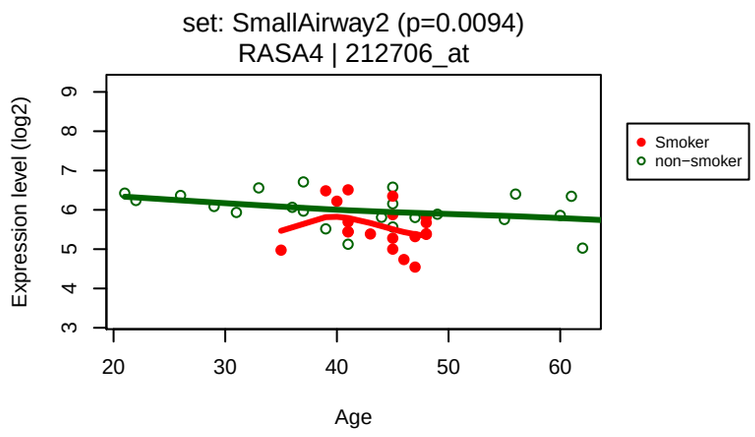
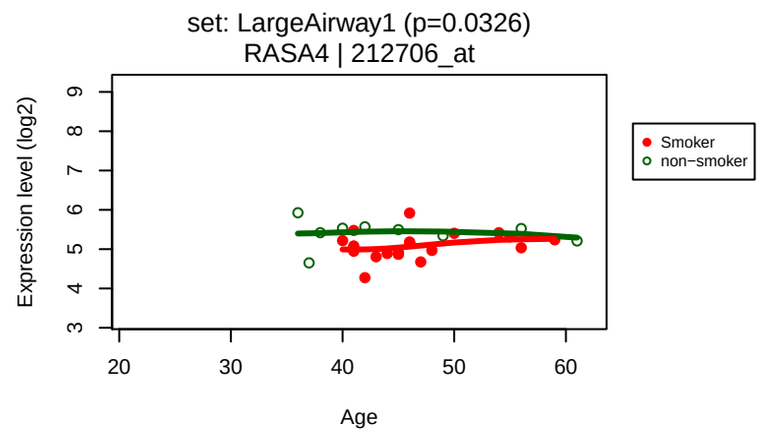

Gene expression of smokers and nonsmokers
gene: RASA4 | 212706_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0273
p=0.0094
p=7e−04
p=0.0326
p=0.2564
p=0.2901
p=7e−04
p=0.411
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

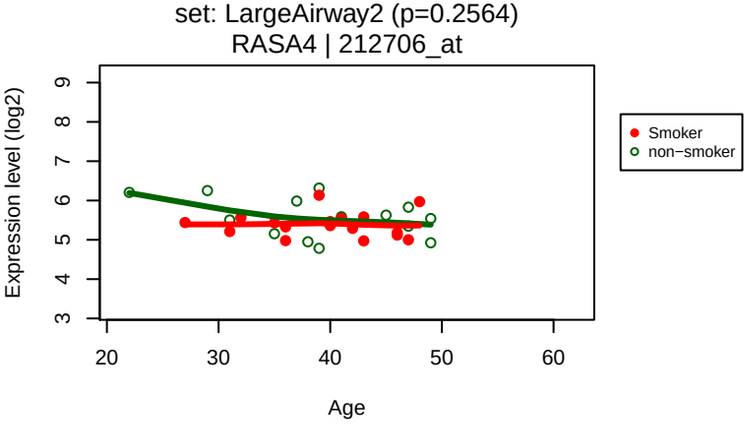

Gene expression ratio (smokers/nonsmokers)
gene: RASA4 | 212706_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0273
p=0.0094
p=7e−04
p=0.0326
p=0.2564
p=0.2901
p=7e−04
p=0.411
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

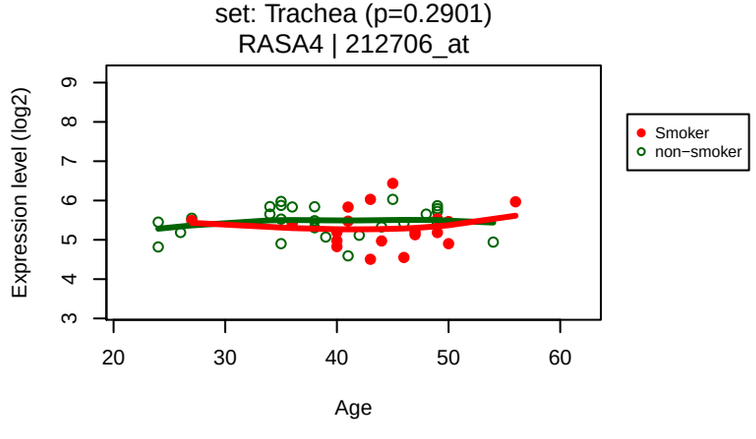

Gene expression ratio in Males
gene: RASA4 | 212706_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0311
p=0.4982
p=0.4576
p=0.3766
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RASA4 | 212706_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1139
p=0.0057
p=0.611
p=0.7795
SmallAirway2
LargeAirway2
Trachea
Macrophage2

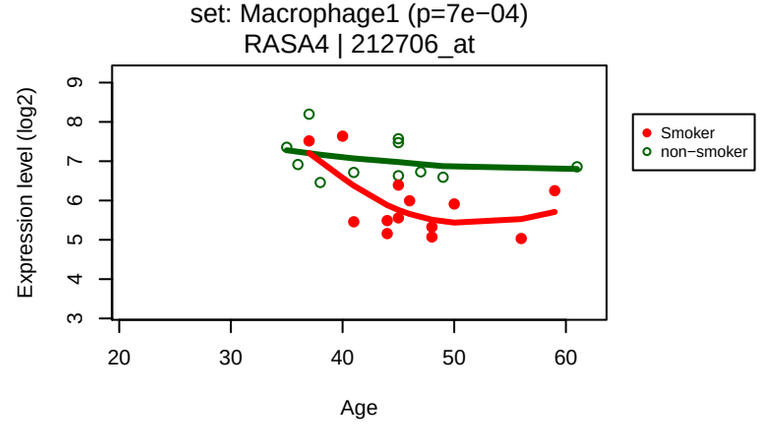

set: SmallAirway1 (p=0.0273)
RASA4 | 212706_at
Expression level (log2)
Age
Smoker
non−smoker

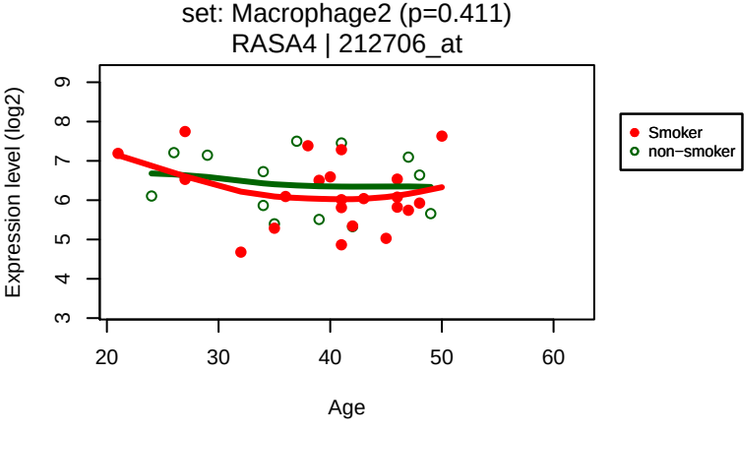

set: SmallAirway2 (p=0.0094)
RASA4 | 212706_at
Expression level (log2)
Age
Smoker
non−smoker

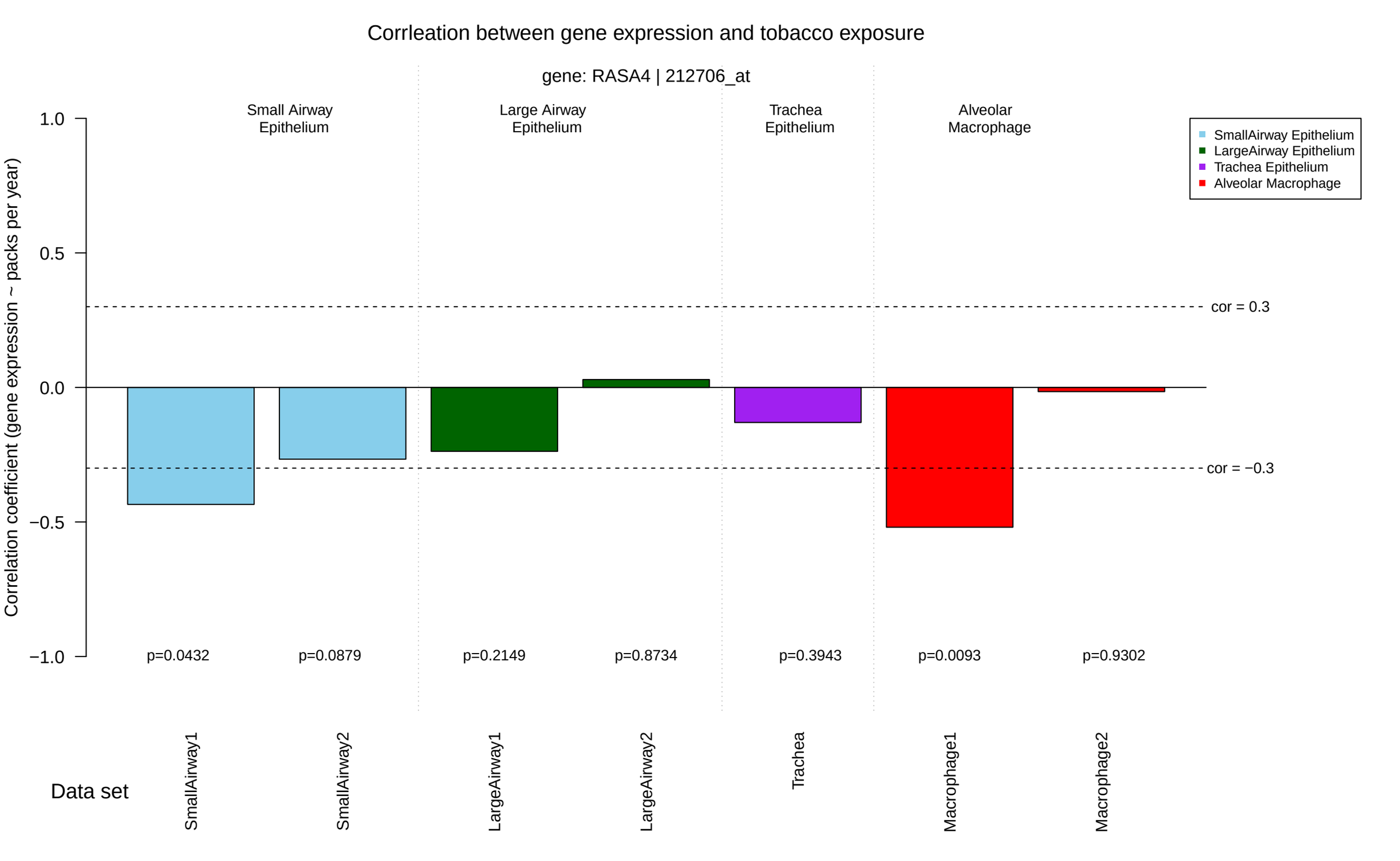

set: LargeAirway1 (p=0.0326)
RASA4 | 212706_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2564)
RASA4 | 212706_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2901)
RASA4 | 212706_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=7e−04)
RASA4 | 212706_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.411)
RASA4 | 212706_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RASA4 | 212706_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0432
p=0.0879
p=0.2149
p=0.8734
p=0.3943
p=0.0093
p=0.9302
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2